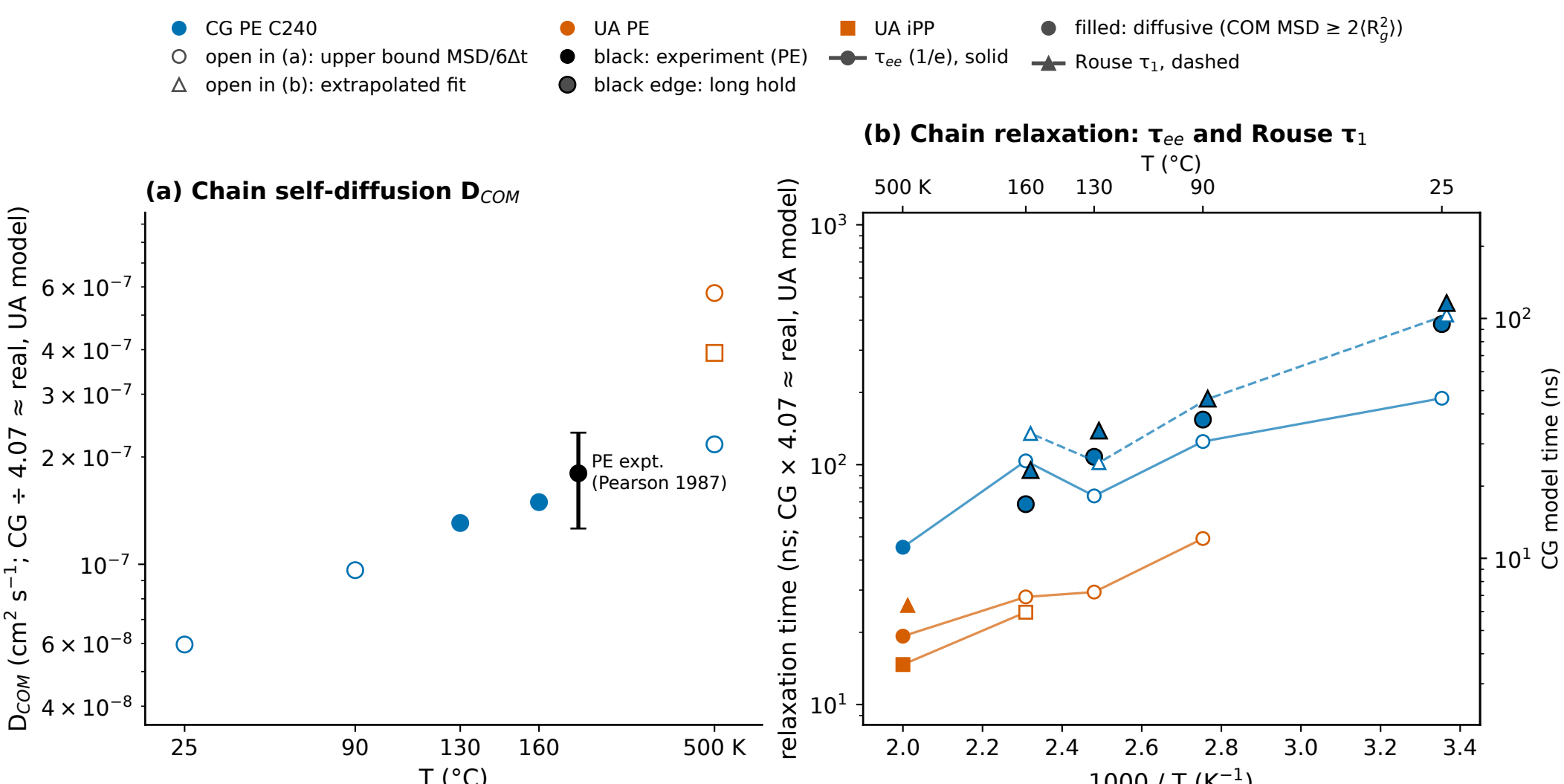
CG PE C240
open in (a): upper bound MSD/6Δt
open in (b): extrapolated fit
UA PE
black: experiment (PE)
black edge: long hold
UA iPP
τ_{ee} (1/e), solid
filled: diffusive (COM MSD ≥ 2⟨R_g^2⟩)
Rouse τ_1, dashed
(a) Chain self-diffusion D_{COM}
D_{COM} (cm^2 s^{-1}; CG ÷ 4.07 ≈ real, UA model)
6×10^{-7}
4×10^{-7}
3×10^{-7}
2×10^{-7}
10^{-7}
6×10^{-8}
4×10^{-8}
PE expt.
(Pearson 1987)
25
90
130
160
500 K
T (°C)
(b) Chain relaxation: τ_{ee} and Rouse τ_1
T (°C)
500 K
160
130
90
25
relaxation time (ns; CG × 4.07 ≈ real, UA model)
10^3
10^2
10^1
CG model time (ns)
10^2
10^1
2.0
2.2
2.4
2.6
2.8
3.0
3.2
3.4
1000 / T (K^{-1})